STRATEGIC ARCHITECTURE & OPERATIONS EXPANSION

# UNIFIED E-COMMERCE ARCHITECTURE & CENTRAL COMMAND PLATFORM.

A comprehensive business roadmap for operating a high-performance consumer online storefront and managing complete catalog, pricing, seasonal offers, and order fulfillment directly from your single CarXen business dashboard.

| PREPARED FOR | INTEGRATION MODEL | PRIMARY ADVANTAGE |
| --- | --- | --- |
| CarXen Leadership & Client Stakeholders | Dual-Application Single Cloud Database Engine | Zero Duplicate Data · 100% Centralized Control |

EXECUTIVE OVERVIEW

# THE UNIFIED RETAIL VISION

Automotive accessories require precision: accurate pricing, guaranteed vehicle fitment, and dependable fulfillment. Managing a separate e-commerce store with disconnected logins inevitably causes stock errors and operational delays. Here is how our single-dashboard solution transforms operations:

### Traditional Disconnected System

- ✕ **Separate Logins:** Staff must switch constantly between an online store admin and the CarXen business software.
- ✕ **Duplicate Effort:** Products, prices, and discounts must be typed in twice across both platforms.
- ✕ **Inventory Desync:** High danger of selling accessories online that were already purchased in-person.
- ✕ **Delayed Dispatch:** Web orders sit unattended in separate mailboxes rather than flowing into the daily work queue.

### CarXen Unified Ecosystem

- ✓ **One Single Command Center:** Manage everything from the dedicated E-Commerce sidebar tab in CarXen.
- ✓ **Instant Live Updates:** Update a price or start an offer in CarXen; the consumer website reflects it in real time.
- ✓ **Unified Inventory Source:** One central stock balance across in-store and online channels prevents overselling.
- ✓ **Live Order Alerts:** Web purchases arrive straight onto your operational dashboard ready for immediate dispatch.

## SYSTEM ARCHITECTURE FLOWCHART: DUAL TOUCHPOINTS, ONE ENGINE

SHARED SUPABASE CLOUD DATABASE

CUSTOMER TOUCHPOINT

### Public Online Storefront

Clean, ultra-fast consumer ecommerce website deployed on high-speed global cloud.

- ✓ Vehicle Fitment Selector
- ✓ Multi-Angle Product 3D Views
- ✓ Instant UPI & Card Checkout

**Supabase Cloud**
Isolated Schemas

ADMIN COMMAND

### CarXen Master Dashboard

Your daily operational software containing the new native "E-Commerce" management hub.

- ✓ Product Studio & Price Controls
- ✓ 1-Click Promotional Offers
- ✓ Live Web Order Stream & Dispatch

## WHY USE ONE CLOUD DATABASE WITH DIFFERENT SCHEMAS?

ZERO COMPLEXITY

By connecting both the public website and the CarXen dashboard to the same Supabase database on secure, isolated schemas, both systems communicate instantaneously. There is no fragile third-party sync tool that can break, no recurring extra subscription costs, and no delay.

DASHBOARD INTEGRATION

# THE E-COMMERCE SIDEBAR MODULE

Within the CarXen business dashboard, an "E-Commerce" section is introduced into the primary left navigation sidebar. It provides complete managerial control over the online storefront without disrupting existing business features:

**CarXen Master Sidebar Navigation**

Core Workshop Modules + **Dedicated E-Commerce Control Hub**

NATIVE DASHBOARD MODULE

### 1. Product Studio

Add new accessories in 60 seconds with photos, pricing, compare rates, and descriptions.

- Stitch Colors
- Stock Toggle
- Specs Table

### 2. Offers & Coupons

Launch flash sales, percentage discounts, and custom promo codes with start/end rules.

- Coupon Codes
- Bundle Deals
- Free Ship Target

### 3. Order Stream

Real-time order pipeline. Advance order status from Confirmed to Packed to Dispatched.

- Live Alerts
- Packing Slips
- AWB Tracking

### 4. Vehicle Fitment

Configure which accessories match specific vehicle makes, models, years, and fuel types.

- Brand Database
- Model Mapping
- Universal Tags

### 5. Sales Analytics

Track daily web sales, average order value, top car accessories, and visitor conversion.

- Daily Revenue
- Top Sellers
- Pincode Demand

### 6. Store Settings

Configure shipping fees, delivery pincodes, store contact numbers, and return policies.

- Courier Rates
- Pincode Limits
- GST Settings

TWO-WAY DATA FLOW

## HOW ACTIONS FLOW BETWEEN DASHBOARD & WEBSITE

STEP 01

**Admin Action**

Owner creates a 10% Diwali offer or adds a new steering cover in CarXen.

→

STEP 02

**Direct Cloud Save**

Supabase updates the central record instantaneously without intermediate sync apps.

→

STEP 03

**Storefront Live**

Online store reflects the new price or product immediately on customer phones.

→

STEP 04

**Order Influx**

Customer purchases; order notifies CarXen staff instantly and updates stock.

OPERATIONAL CONTROL

# FULL CLIENT CAPABILITIES MATRIX

This matrix details every practical feature the client gains to manage, market, and fulfill online car accessory sales with total autonomy:

| FEATURE MODULE | WHAT THE CLIENT CAN HANDLE | OPERATIONAL VALUE |
|---|---|---|
| **Product & Catalog Studio**<br>Item Creation & Editing | Upload photos, edit descriptions, adjust retail prices, set strike-through discounts, manage stitch colour variants, and toggle items out-of-stock instantly. | `LIVE IN 60S`<br>No web developer needed for item updates. |
| **Offer & Promotion Engine**<br>Discounts & Coupons | Create custom coupon codes (e.g., `CARXEN10`, festive codes), configure minimum cart amounts for free shipping, and launch "Bought Together" bundle deals. | `1-CLICK LAUNCH`<br>Drives higher Average Order Value. |
| **Order Stream & Dispatch**<br>Fulfillment Pipeline | View live web orders with vehicle details, customer address, and items. Update status: *Confirmed* ➔ *Packing* ➔ *Dispatched* ➔ *Delivered*. Print GST slips. | `AUTOMATED`<br>Eliminates missed orders or shipping errors. |
| **Vehicle Fitment Matcher**<br>Compatibility Engine | Assign products to specific cars (Creta, Thar, Nexon, etc.) or mark as Universal. When customers pick their car online, only verified fitting parts are presented. | `ZERO RETURNS`<br>Drastically cuts return and exchange rates. |
| **Review Moderation**<br>Social Proof & Ratings | Read customer reviews and star ratings before they appear on product pages. Approve authentic buyer reviews with one click to build brand credibility. | `BRAND CONTROL`<br>Protects brand reputation online. |
| **Storefront Analytics**<br>Performance Metrics | Understand revenue breakdown, highest-demand vehicle models, top-performing product categories, and geographical sales distribution across states. | `REAL-TIME`<br>Actionable data to stock what actually sells. |

CUSTOMER TO DELIVERY

## THE COMPLETE ORDER LIFECYCLE FLOWCHART

`STAGE 1: PLACED`
**Customer Checkout**
Customer selects car, adds quilted seat covers, and pays via UPI/Card.

→

`STAGE 2: DASHBOARD`
**Staff Notification**
Order appears on CarXen dashboard with exact vehicle spec and address.

→

`STAGE 3: PACKING`
**Slip & Dispatch**
Staff clicks 'Print Packing Slip', hands box to courier, and enters AWB.

→

`STAGE 4: COMPLETE`
**Customer Updates**
Customer gets live SMS/WhatsApp tracking; stock balance deducts automatically.

IMPLEMENTATION PLAN

# PHASED ROLLOUT ROADMAP

We execute the upgrade in four structured phases to guarantee zero disruption to your daily workshop business operations:

PHASE 01

### Schema & Architecture

- Create isolated ecommerce schema inside existing Supabase.
- Set up tables for products, vehicle fitment, discounts, and orders.
- Establish secure cloud API connections.

PHASE 02

### Storefront Deploy

- Deploy high-speed Next.js consumer ecommerce website.
- Connect dynamic product catalog and car fitment selectors.
- Integrate UPI/Card payment gateway.

PHASE 03

### Dashboard Module

- Build the dedicated "E-Commerce" left sidebar tab in CarXen.
- Integrate Product Studio, Offer Manager, and Fitment Matrix.
- Enable live order alerts & status workflow.

PHASE 04

### Testing & Launch

- Execute test orders from cart to invoice generation.
- Staff walkthrough on processing orders and adding products.
- Official public commercial launch.

ENTERPRISE RELIABILITY

## SECURITY & DATA ISOLATION GUARANTEES

### Airtight Security & Data Partitioning

The public storefront only possesses permission to read public catalog items and write new customer orders. It has zero access to your internal business financial records, workshop accounts, or supplier invoices.

### Zero Maintenance Burden

Because both the storefront and the dashboard run off your primary cloud database, you only have one database setup. Backups, cloud security, and maintenance happen automatically in the background.

## CONCLUSION: THE CARXEN ADVANTAGE

EXECUTIVE TAKEAWAY

By introducing the **E-Commerce Module** directly inside the CarXen business dashboard, you achieve the best of both worlds: a world-class, dedicated online store that wows automotive customers, paired with the supreme convenience of managing inventory, promotions, and dispatches from the single system your team already uses every day.